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Dipartimento di
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Electronics for Embedded Systems

Department of Electronics and Telecommunications (DET)
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Experiment Title:	Lab 5 – Transistors in switching mode
Course:	Electronics for Embedded Systems
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Date of Experiment:	13/01/2026
Date of Submission:	25/01/2026
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Abstract

In this laboratory experiment, we explored the application of a Bipolar Junction Transistor (BJT 2N3700) as a solid-state switch controlled by an FPGA (DE1-SoC) to drive a power load (LED). The main objectives were to generate a Pulse Width Modulation (PWM) signal, dimension the base resistor (R_B) to ensure transistor saturation while complying with the FPGA's current limits, and analyze the switching transients. We measured the PWM characteristics at 10 kHz and 640 Hz. For the BJT circuit, with a calculated R_B of 180Ω , we observed a significant storage time ($t_{\text{delay_rise}} \approx 11\mu\text{s}$) in the standard configuration due to deep saturation. To improve switching performance, we tested a Baker Clamp (using a 1N4148 diode) and a Speed-Up Capacitor (100 nF). Results indicated that the speed-up capacitor offered the best performance, drastically reducing the turn-off delay to approximately $0.05\mu\text{s}$ by actively removing stored base charges.

1. Introduction

Digital circuits, such as FPGAs, typically operate at low voltages (3.3V) and have limited output current capabilities (e.g., 16 mA max for the DE1-SoC GPIO pins), making them unsuitable for directly driving high-power loads like motors or high-brightness LEDs. To interface these devices, a Bipolar Junction Transistor (BJT) can be used as a switch.

The goals of this laboratory are:

1. **PWM Implementation:** To design and verify a PWM generator using Verilog on the DE1-SoC, capable of variable duty cycles and frequencies.
2. **Circuit Sizing:** To calculate the appropriate base resistance R_B that guarantees BJT saturation ($V_{CE} \approx 0.2V$) without exceeding the maximum current rating of the FPGA pins.
3. **Transient Analysis:** To measure the switching times (t_{rise} , t_{fall}) and propagation delays, specifically the storage time caused by minority carrier accumulation in the base region.
4. **Optimization:** To compare two methods for reducing switching delays: the Baker Clamp circuit (using a diode) and the speed-up capacitor technique.

2. Materials and Methods

Hardware Setup:

- **FPGA Board:** Terasic DE1-SoC (Cyclone V).
- **Transistor:** NPN BJT 2N3700.
- **Load:** 5mm Green LED.
- **Components:**
 - Collector Resistor (R_C): 560Ω (initially calculated as 470Ω , adjusted to 560Ω in the final setup).
 - Base Resistor (R_B): 180Ω .

- Diode: 1N4148 (Fast switching diode for Baker Clamp).
- Capacitor: 100 nF (Ceramic, marking '104', for Speed-Up).
- **Instruments:** Rigol DS1054Z Oscilloscope, Digital Multimeter, Power Supply 12V.

Methodology:

1. **PWM Generation:** A PWM module was synthesized on the FPGA. We verified the output frequency (f_{PWM}) and duty cycles (20%, 50%, 100%) for two configurations: high frequency (~ 10 kHz) and low frequency (~ 640 Hz).
2. **BJT Switch Design:** We calculated the limits for R_B based on $V_{OH}=3.3V$, $V_{BE}=0.7V$, and $I_{MAX}=16mA$. A value close to the minimum was selected to force deep saturation.
3. **Measurements:**
 - Static voltages (V_H , $V_{CE(sat)}$) were measured with a multimeter.
 - Dynamic switching parameters (t_{rise} , t_{fall} , t_{delay}) were captured using the oscilloscope cursors.
 - The circuit was tested in three configurations: "Nothing" (Resistor only), "With Diode" (Baker Clamp), and "With Capacitor" (Speed-Up).

3. Results

3.1 PWM Signal Verification

The PWM generator was tested with MAXCNT=10 (high freq) and MAXCNT=98 (low freq). The measured values matched the theoretical expectations. A visual comparison of the LED brightness at different duty cycles is provided in **Figure 1&2**.

12	Divisor	MaxCnt	REG	f_{PWM}	T_{ON}	T_{OFF}	duty	V_H	V_L
20%	29	10	2	9.77kHz	20.5 μ s	82 μ s	18.05%	3.73	0
50%	29	10	5	9.78kHz	51.1 μ s	51.1 μ s	45.40%	3.76	0
100%	29	10	11	10kHz	-	-	100%	3.76	0

Table 1

100	Divisor	MaxCnt	REG	f_{PWM}	T_{ON}	T_{OFF}	duty	V_H	V_L
20%	3	98	20	645Hz	310 μ s	1240 μ s	20%	3.72	0
50%	3	98	50	641Hz	780 μ s	780 μ s	50.64%	3.72	0
100%	3	98	98	643Hz	1539.9 μ s	15.08 μ s	99.03%	3.72	0

Table 2

3.2 BJT Circuit Design and DC Analysis

The base resistance R_B was dimensioned to protect the FPGA ($I_{OH} \leq 16\text{mA}$) and ensure saturation ($I_B \gg I_C/h_{FE}$)

- **Calculated R_{Bmin} :** $\frac{3.3V - 0.7V}{16mA} = 162.5\Omega$,
- **Calculated R_{Bmax} :** $\frac{3.3V - 0.7}{2mA} = 1300\Omega$,
- **Selected R_B :** 180Ω .

DC Measurements: With the circuit loaded (LED ON), the following values were recorded:

- **V_H (FPGA Output):** 2.4 V (Note the drop from the unloaded 3.76 V).
- **$V_{CE(sat)}$:** 11.8 mV (Deep saturation).
- **$V_{CE(cut-off)}$:** 10.5 V.
- **Power Dissipation R_C :** 171 mW.
- **Power Dissipation R_B :** 14 mW.

3.3 Switching Time Analysis

The dynamic performance was evaluated in three configurations. **Figure 3** illustrates the typical switching waveforms and the delay measurements captured on the oscilloscope. The results highlight the impact of storage time.

Time	Nothing	1N4148	C = 100nF
t_{rise}	2 μs	1,5 μs	0,1 μs
t_{fall}	0,1 μs	0,1 μs	0,05 μs
t_{delay_rise}	11 μs	0,5 μs	0,05 μs
T_{delay_fall}	0,05 μs	0,05 μs	0,02 μs

4. Discussion

4.1 Loading Effect on V_H

In the PWM experiments (unloaded), V_H was measured around 3.7 V. However, when connected to the BJT base with $R_B = 180\Omega$, V_H dropped to 2.4 V. This voltage sag is due to the internal impedance of the FPGA's GPIO driver. With $R_B = 180\Omega$ the base current is substantial ($I_B \approx 9.8\text{mA}$), causing a significant voltage drop across the FPGA's internal PMOS output transistor. Despite this, 2.4 V is sufficient to drive the BJT.

4.2 Analysis of Delays

- Standard Circuit ("Nothing"): The most critical observation is the Turn-OFF delay of 11 μs (**Figure 3**). This is the Storage Time. Since we designed the circuit for hard saturation ($I_B \gg I_{C(sat)}/\beta$), excess minority carriers accumulate in the base region.

When the base drive is removed, the transistor continues to conduct until these carriers recombine, causing a long delay before V_{CE} rises.

- **Baker Clamp (1N4148)**: Adding a diode between Base and Collector prevents the transistor from entering deep saturation by clamping V_{CE} so it cannot drop significantly below V_{BE} . This reduced the stored charge, dropping the delay from $11\mu s$ to $0.5\mu s$, improving switching speed without complex circuitry.
- **Speed-Up Capacitor (100nF)**: Placing a capacitor in parallel with R_B created a "differentiation" effect. During turn-on, the capacitor acts as a short, providing a high current spike (overdrive) to turn on the BJT instantly. During turn-off, it provides a negative current spike, actively extracting the stored charges from the base. This resulted in the fastest performance, with negligible delay ($0.05\mu s$) and sharper edges.

5. Conclusions

The experiment successfully demonstrated the design trade-offs in BJT switching circuits. While a low base resistance ($180\ \Omega$) ensures robust saturation ($V_{CE} \approx 12mV$), it introduces a significant storage delay ($11\ \mu s$) that limits switching frequency.

The comparison of optimization techniques confirmed that:

1. The **Baker Clamp** is effective for reducing storage time significantly (down to $0.5\ \mu s$) while maintaining a simple topology.
2. The **Speed-Up Capacitor** is the superior solution for high-speed applications, virtually eliminating both storage delay and transition times ($<0.1\ \mu s$), making the BJT an ideal switch for the generated PWM signals.

6. References

[1] Laboratory Manual, "Lab 5: BJT Switch and PWM", Electronics for Embedded Systems, Politecnico di Torino. [2] 2N3700 Datasheet. [3] Altera DE1-SoC User Manual (GPIO Current Strength Tables).

Appendix A – Figures and Tables

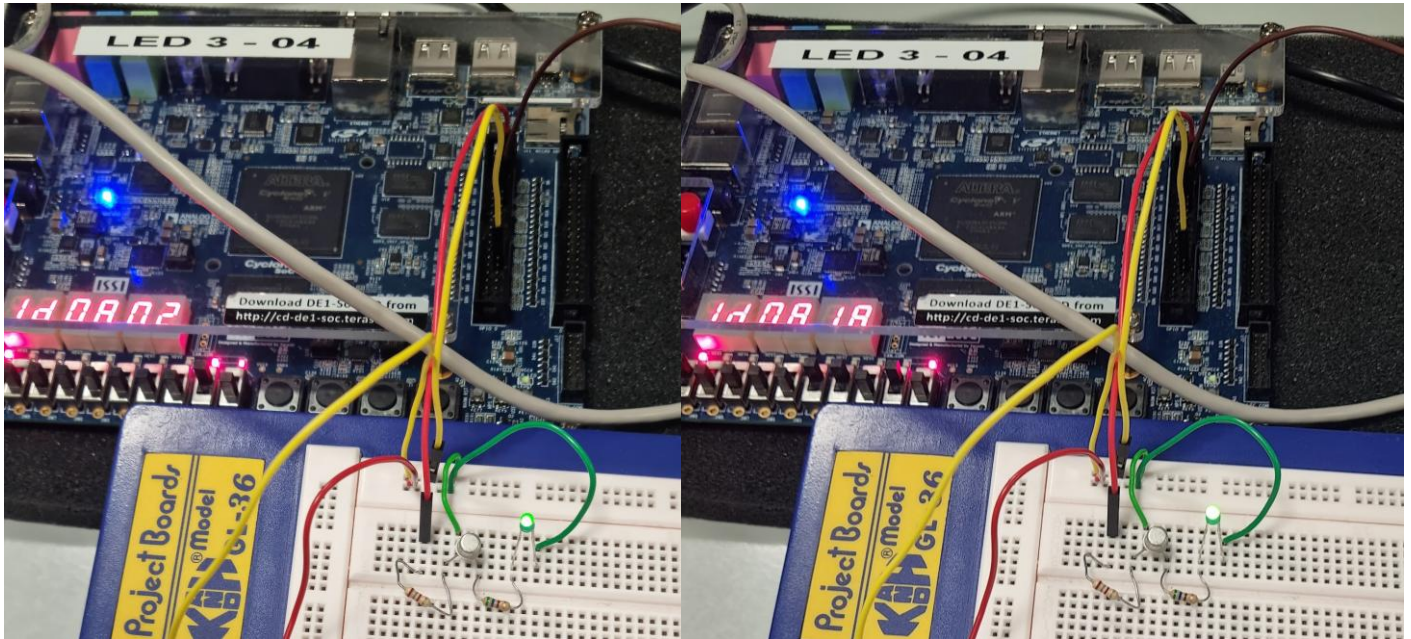


Figure 1 & 2: Visual comparison of LED brightness modulation controlled by the FPGA. The image on the left demonstrates the LED operating at a low PWM duty cycle (approx. 20%), resulting in lower luminous intensity. The image on the right shows the same circuit driven with a higher duty cycle, resulting in significantly higher brightness due to the increased average current flowing through the load. The 7-segment displays on the FPGA board indicate the different register settings used to vary the pulse width.

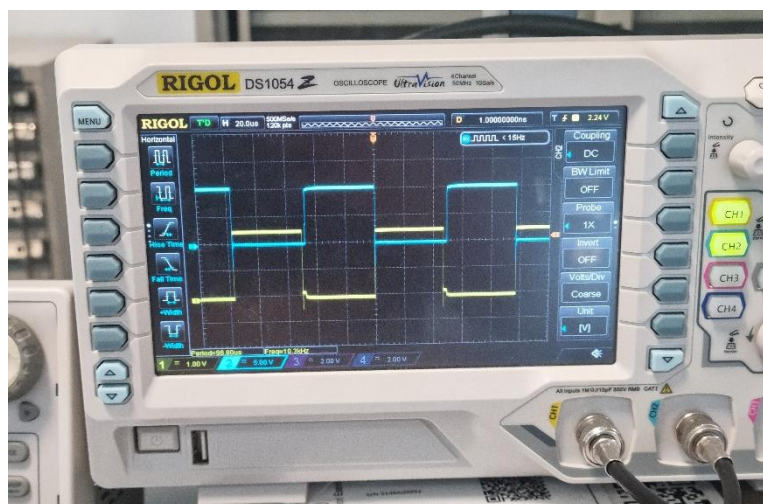


Figure 3: Oscilloscope capture of the BJT switching waveforms.

The image displays the dynamic behaviour of the transistor in the Common Emitter configuration.

- Channel 1 (Yellow): Shows the PWM control signal applied to the Base (V_B). This is the driving signal generated by the FPGA.
- Channel 2 (Blue): Shows the Collector-Emitter voltage (V_{CE}).

The waveforms demonstrate the inverting nature of the switch: when the base signal is High, the transistor saturates, pulling the collector voltage to near 0V. Conversely, when the base signal is Low, the transistor is in cut-off, and the collector voltage rises to the supply voltage ($V_{CC}= 12V$).